

Group: _____ **Section:** _____

Member 1: _____ **BUID:** _____

Member 2: _____ **BUID:** _____

Member 3: _____ **BUID:** _____

Member 4: _____ **BUID:** _____

Project 1 Modeling, Model Selection, and Diagnostics

Purpose

Today you will fit candidate Project 1 models, compare them, diagnose model issues, and decide what evidence supports your final modeling strategy.

Shared Group Workflow

Work in one shared Google Doc and one shared analysis folder. Assign the four roles below, then have every member inspect the models and sign off on the interpretation.

- **Member 1 – Modeling lead:** fit and document the candidate models.
- **Member 2 – Comparison lead:** compare fit, prediction, and interpretability evidence.
- **Member 3 – Diagnostics lead:** check assumptions, influential observations, and remedial options.
- **Member 4 – Interpretation editor:** connect results to the research question and maintain the shared report.

In-Class Work

- (A) **Candidate models.** Fit linear, multiple regression, or logistic regression models as appropriate for your response variable.
- (B) **Model comparison.** Compare candidate models using summaries such as adjusted R^2 , AIC, residual behavior, prediction quality, or interpretability.
- (C) **Diagnostics.** Check residual plots, influential observations, transformations, or classification summaries as appropriate.
- (D) **Interpretation.** Translate key coefficients or predictions into language connected to your research question.
- (E) **Check-in.** Which model is currently strongest, and what concern still needs to be addressed?

Deliverable

Submit the Project 1 progress report through Gradescope or Blackboard by Sunday, October 4 at 10:00 PM. Include candidate models, diagnostics, a justified modeling direction, and the accessible shared Google Doc link.